Chapter 3: Key Theories and Models of Behaviour Change Communication: Individual Level

1.0 Introduction to Individual-Level Behaviour Change

The discipline of Social and Behaviour Change Communication (SBCC) operates on the fundamental empirical premise that the mere dissemination of information is rarely sufficient to alter entrenched human behaviours. Historically, early public health and development campaigns relied heavily on the "knowledge-deficit" model, assuming that if individuals were simply educated about a risk, they would rationally and automatically adjust their actions to mitigate that risk.

Decades of public health data have disproved this assumption, revealing a profound gap between knowledge, attitude, and actual behavioural execution. To bridge this gap and design effective interventions, practitioners and researchers must understand the complex cognitive, emotional, and environmental architectures that dictate human decision-making.

Within the broader socio-ecological framework, which examines influences across public policy, community, institutional, and interpersonal levels, individual-level theories of behaviour change form the foundational bedrock.

These theories map the internal cognitive processes of the individual, conceptualising human behaviour not as a random or purely instinctual occurrence, but as a systematic output generated by specific belief structures, risk appraisals, self-efficacy levels, and environmental triggers.

Individual-level models attempt to isolate the psychological variables that act as barriers or facilitators to change, providing SBCC strategists with a diagnostic toolkit to identify precisely why a target population behaves the way it does, and a prescriptive blueprint for how to alter that trajectory.

This chapter systematically examines the most prominent individual-level theories and models utilised in contemporary SBCC. It explores the Health Belief Model (HBM), the Theory of Planned Behaviour (TPB) and Reasoned Action (TRA), the Stages of Change or Transtheoretical Model (TTM), and Social Cognitive Theory (SCT).

Furthermore, it investigates other highly relevant frameworks, namely Protection Motivation Theory (PMT) and the Fogg Behaviour Model (FBM), which have gained significant traction in the era of digital health interventions. By dissecting the core constructs of each framework and rigorously examining their application in empirical case studies drawn from global academic literature, this chapter aims to demonstrate how these psychological architectures are operationalised in real-world public health, environmental, and social interventions.

2.0 The Health Belief Model (HBM)

The Health Belief Model (HBM) remains one of the most foundational, heavily researched, and widely applied frameworks in health behaviour research and communication. Conceptualised in the 1950s by social psychologists operating within the United States Public Health Service (USPHS), the model was originally developed to investigate a pressing programmatic failure: the widespread refusal of individuals to adopt disease preventatives or participate in free screening tests for the early detection of asymptomatic conditions, such as tuberculosis. The researchers realised that removing financial and physical barriers was insufficient; psychological barriers were the primary deterrents.

2.1 Core Cognitive Constructs

The HBM is fundamentally an expectancy-value theory. It posits that an *individual's readiness to act regarding a specific health condition is determined by a rational assessment of the perceived threat and the perceived utility of the recommended behaviour.*

The model relies on six primary cognitive dimensions that must be addressed in any SBCC campaign:

Construct	Conceptual Definition	SBCC Behavioural Implication
Perceived Susceptibility	An individual's subjective assessment of their personal risk of developing a health condition or contracting a disease.	Messages must personalise the risk to the target audience. General statistics often fail to elevate personal susceptibility.
Perceived Severity	The individual's evaluation of the seriousness of the condition and its potential medical, financial, or social consequences.	Interventions must vividly illustrate the tangible impacts of the disease to ensure the threat is not minimised.
Perceived Benefits	The belief in the efficacy of the advised action to successfully reduce the risk or seriousness of the impact.	Communication must clearly demonstrate that the recommended action yields tangible, achievable advantages.
Perceived Barriers	The individual's assessment of the tangible and psychological costs of the advised action (e.g., expense, pain, stigma, inconvenience).	If perceived barriers outweigh perceived benefits, the behaviour will not occur. SBCC must actively reduce cognitive friction.
Cues to Action	External or internal stimuli that trigger the decision-making process (e.g., a physician's advice, bodily symptoms, mass media campaigns).	Strategic communication serves as the direct external cue to prompt the initiation of the targeted behaviour.
Self-Efficacy	The personal confidence in one's own ability to successfully execute the behaviour required to produce the desired outcomes.	High self-efficacy is required to overcome complex barriers, especially for long-term chronic disease management.

While foundational, the HBM is not without theoretical limitations. Critics argue that the framework overly emphasises cognitive constructs and assumes highly rational decision-making, thereby neglecting the emotional complexities, cultural nuances, and socio-economic realities that influence human health behaviours.

Traditional applications of the HBM have also been criticised for treating the constructs as static, parallel variables, which can result in limited predictive power—sometimes as low as 20% to 40% variance explained when compared to ecological models that aggressively incorporate environmental factors.

2.2 Empirical Case Studies in SBCC

Case Study 1: Variable Ordering in Flu Vaccination Campaigns

The complexity of the HBM's construct interactions was elegantly demonstrated in a large-scale survey of 1,377 adults following an eight-month influenza vaccination SBCC campaign. The study aimed to address theoretical gaps regarding the undefined variable ordering within the HBM—specifically, whether the constructs operate in parallel, in sequence, or are moderated by other variables. Statistical evaluations supported a highly complex model of moderated mediation.

The data revealed that exposure to the communication campaign positively influenced vaccination behaviour, but this effect was indirect. The effect was mediated through perceived barriers and perceived threat, and crucially, this entire pathway was moderated by the individual's level of self-efficacy.

Furthermore, perceived barriers and perceived benefits formed a serial mediation chain. The profound implication for SBCC practitioners is that campaign variables cannot merely be thrown at an audience simultaneously. Interventions must often first build an individual's self-efficacy to ensure that subsequent messages regarding threats and barriers are processed effectively and translated into action. [https://pmc.ncbi.nlm.nih.gov/articles/PMC4530978/]

Case Study 2: Self-Care Behaviours in Type 2 Diabetes

A rigorous cross-sectional survey conducted in 2024 investigated the impact of HBM constructs on self-care behaviours among 1,225 Black/African American men diagnosed with Type 2 Diabetes (T2D). Utilising multiple linear regression adjusted for demographic factors such as age, education, and income, the research generated critical insights into chronic disease management.

The study demonstrated that self-efficacy was the absolute strongest overarching predictor of self-care behaviours across all measured domains ($\beta = 0.055$, $p <$

0.001). Perceived susceptibility was significantly and positively associated with specific tasks, such as healthful eating ($\beta = 0.042$, $p = 0.038$), glucose monitoring ($\beta = 0.117$, $p < 0.001$), and foot care ($\beta = 0.140$, $p < 0.001$).

Perceived severity also positively linked to diet adherence and glucose monitoring. However, the data revealed a striking contradiction to traditional assumptions: perceived benefits did not significantly impact overall self-care behaviours ($\beta = 0.001$, $p = 0.893$). This suggests a saturation of knowledge; for this specific demographic living with a chronic disease, merely reiterating the benefits of management is ineffective. Instead, SBCC interventions must pivot entirely toward enhancing self-efficacy, mitigating perceived barriers ($\beta = -0.047$, $p < 0.001$) and maintaining risk awareness. [https://doi.org/10.3390/ijerph22030414]

Case Study 3: Hypertension Management Education-

The clinical efficacy of HBM-based health education was documented in a study evaluating blood pressure control and medication adherence over three- and six-month intervals in hypertensive patients.

By systematically addressing perceived susceptibility to cardiovascular events (such as stroke), highlighting the severity of uncontrolled hypertension, and practically mitigating perceived barriers to daily medication adherence, the targeted educational intervention led to significantly better physiological health outcomes.

The study reinforced the model's value in designing effective health interventions, while noting that future research must continually adapt the constructs to account for cultural applicability and complex environmental influences. [https://www.wjgnet.com/2307-8960/full/v12/i33/6660.htm]

3.0 Theory of Planned Behaviour (TPB) and Reasoned Action (TRA)

The Theory of Reasoned Action (TRA) and its subsequent, more robust iteration, the Theory of Planned Behaviour (TPB), represent a highly structured framework used to understand, predict, and systematically influence human behaviour.

Developed originally by Martin Fishbein and Icek Ajzen, these models pivot away from the threat-based architecture of the HBM and instead posit that the most proximal, immediate determinant of behaviour is the individual's intention to perform it.

3.1 Core Mechanisms and Belief Structures

The TPB expands upon the TRA by formally adding the construct of perceived behavioural control, thereby acknowledging the reality that individuals do not always possess complete volitional control over their actions due to environmental constraints or lack of resources. The framework is built upon four primary, interconnected belief structures that drive specific model constructs, which ultimately dictate behavioural intention:

1. **Attitude toward the Behaviour**: Driven by underlying *behavioural beliefs*, this encompasses an individual's positive or negative evaluation of performing the specific behaviour. Importantly, this includes both cognitive assessments (rational logic) and affective beliefs (anticipated emotional responses or feelings associated with the action).

2. **Subjective Norms**: Driven by *normative beliefs*, this construct captures the perceived social pressure or support regarding the behaviour. It reflects how much an individual believes that significant others, such as family members, peers, religious leaders, or healthcare providers, think they should or should not perform the behaviour, combined with the individual's motivation to comply with those referents.

3. **Perceived Behavioural Control (PBC)**: Driven by *control beliefs* and *efficacy beliefs*, this construct relates to an individual's perception of the specific factors that facilitate or hinder the execution of a behaviour. It functions similarly to self-efficacy and is often examined alongside locus of control.

4. **Behavioural Intention**: The central determinant of action in the TPB, shaped by the combined, weighted influence of Attitude, Subjective Norms, and PBC. Generally, a positive attitude coupled with positive subjective norms results in greater perceived control, synergistically increasing the likelihood that intentions will govern actual behavioural changes.

A critical methodological requirement of applying the TPB in SBCC is the process of "salient belief elicitation". Researchers cannot assume they know what drives a population. They must elicit specific content for these measures directly from the target population regarding the behaviour under investigation. This ensures that the targeted communication messages accurately reflect the specific normative, behavioural, and control beliefs of that unique demographic.

3.2 Empirical Case Studies in SBCC

Case Study 1: The SIPsmartER Programme in Rural Populations

The SIPsmartER programme stands as a premier example of TPB translation into community health promotion. Implemented in medically underserved, rural counties of southwest Virginia, the intervention specifically targeted English-speaking adults consuming at least 200 kilocalories of sugar-sweetened beverages (SSBs) daily.

The programme utilised small-group educational classes and "teach-back" methods, a safety communication technique where participants must explain back the health information in plain language to confirm accurate comprehension.

Evaluated via a randomised controlled trial against a comparison group, SIPsmartER proved highly effective. Participants demonstrated a significantly greater decrease in caloric consumption of SSBs ($p < 0.001$), fluid ounce consumption ($p < 0.01$), and total daily energy intake (-277 kcals vs. -43 kcals, $p < 0.01$). The intervention also resulted in positive physical health metrics, including decreases in weight and BMI.

Most importantly for behavioural science, the programme successfully shifted the underlying TPB constructs: the SIPsmartER group exhibited significantly greater increases in scores for SSB attitudes (0.7 vs. 0.2, $p < 0.001$), perceived behavioural control (0.6 vs. 0.3, $p < 0.05$), and behavioural intentions (1.0 vs. 0.3, $p < 0.001$). The programme achieved a Reach of 80% and Effectiveness of 66.7% on the RE-AIM evaluation framework. [https://www.ruralhealthinfo.org/toolkits/health-promotion/2/theories-and-models/reasoned-action]

Case Study 2: Cultural Adaptation of TPB in Kibera, Kenya

While the TPB was originally designed for application within Western cultures, formative field research conducted in Kibera, Kenya - Africa's largest urban informal settlement—demonstrated its profound adaptability for global health interventions when viewed through a rigorous cultural lens.

The study aimed to design an intervention to reduce childhood diarrhoea by promoting the consistent use of point-of-use Safe Water Systems (SWS).

The research revealed that applying the TPB was entirely feasible, provided that normative behaviours and beliefs were assessed strictly through Kenya's societal norms, which feature a collectivist culture and a high-context communication style. To appropriately address the "Subjective Norm" construct in a society where community pressure outweighs individual intention, the intervention was

structured around a women's group train-the-trainer programme. By utilising peer education, behavioural modelling, and motivational interviewing principles, positive normative pressures were engineered directly into the community fabric, effectively leveraging the collectivist culture to drive the TPB framework. [https://digitalcommons.du.edu/cgi/viewcontent.cgi?article=1193&context=etd]

Case Study 3: Climate Risk Communication for Children

The Theory of Reasoned Action was applied to a completely different context: predicting and influencing climate change adaptation behaviours among children in Can Tho city, Vietnam, an area highly vulnerable to climate impacts.

A randomised field experiment utilising a 2x2 design explored the differences between pre- and post-communication treatments. The study confirmed that TRA was a significant predictor model for children's climate change behaviour intentions. Specifically, attitude had a significant independent effect on the intention to act, whereas video treatments focusing solely on subjective norms did not.

However, the statistical interaction of both constructs significantly influenced intentions. The theory-inspired communication strategy successfully altered salient beliefs regarding both attitudes and normative social norms, demonstrating the framework's versatility across age groups and subject matters. [https://doi.org/10.3390/su10062019]

4.0 The Stages of Change / Transtheoretical Model (TTM)

The Transtheoretical Model (TTM), more commonly referred to in public health as the Stages of Change model, revolutionised behavioural science by introducing a chronological, stepwise framework for understanding intentional human behaviour change.

Unlike the TPB or HBM, which generally treat intention as a static or binary variable, the TTM recognises that behaviour change is a temporal process. Individuals move through distinct phases of cognitive readiness, and crucially, interventions must be meticulously matched to the individual's current stage to be effective.

4.1 Core Constructs: Stages and Processes of Change

Developed extensively by James Prochaska and Carlo DiClemente through early work in smoking cessation, the TTM is built upon specific stages of readiness, accompanied by ten psychological and behavioural "processes of change" that

facilitate movement between these stages. Furthermore, progression is governed by "Decisional Balance" (weighing the pros and cons) and an increase in self-efficacy to resist temptations.

Stage of Change	**Characteristics of the Individual**	**Relevant Processes of Change**
Precontemplation	Unaware of or in denial about the problem. No intention to seek a solution or change behaviour in the foreseeable future (next 6 months). Often defensive, resistant, and fixated on the negative aspects (costs) of change.	**Consciousness-raising** (learning new facts); **Environmental re-evaluation** (noticing how the behaviour impacts others).
Contemplation	Aware of the problem and actively considering change within the next 6 months. Experiencing deep ambivalence as they weigh the pros and cons.	**Self-re-evaluation** (assessing self-image with and without the behaviour); **Dramatic relief** (experiencing negative emotions regarding the risk).
Preparation	Intending to take action in the immediate future (next 30 days). The individual has typically taken some significant behavioural step already, such as consulting a physician or buying a gym membership.	**Self-liberation** (making a firm commitment or decision to change).
Action	Actively modifying behaviour, environment, or experiences to overcome the problem. This stage requires significant time, energy, and commitment.	**Counterconditioning** (substituting healthy behaviours); **Stimulus control** (removing triggers); **Reinforcement management** (rewarding positive steps).
Maintenance	Sustaining the behaviour change over time (usually defined as 6+ months) and actively working to prevent relapse into old habits.	**Helping relationships** (relying on social support); Continued **Stimulus control**.

Basic research across 12 different health behaviours has generated a vital rule of thumb for SBCC practitioners regarding at-risk populations: approximately 40% are in the precontemplation stage, 40% are in contemplation, and only 20% are in the preparation stage. This data implies that an action-oriented SBCC campaign that fails to segment its audience will likely alienate 80% of the target population who are not yet cognitively ready to act.

4.2 Empirical Case Studies in SBCC

Case Study 1: Smoking Cessation Interventions and Stage Distribution

The efficacy of the TTM in structuring smoking cessation interventions is heavily documented. In a representative study of 2,422 Korean adult smokers, demographic mapping validated the rule of thumb: 37.6% of the population were in precontemplation, 56.0% in contemplation, and only 6.4% in preparation.

Age-stratified analyses revealed that socioeconomic characteristics—such as education, residence, and household income—along with smoking onset age, significantly influenced an individual's stage of change. This suggests that demographic targeting is essential for effective stage-matched communication.

Similarly, research on South Korean adolescents found that those who lacked knowledge regarding their father's educational level had a significantly lower probability of reaching the preparation stage, whereas higher academic performance generally correlated with movement toward the preparation stage.

An intervention promoting workplace smoking cessation in Taiwan operationalised the TTM processes directly through tailored actions. For individuals in the contemplation stage, the SBCC strategy actively utilised *self-re-evaluation*, encouraging participants to assess their feelings about uncontrolled smoking and its potential long-term illnesses.

For those progressing into the preparation and action stages, the intervention shifted abruptly. It deployed *stimulus control*, asking participants to rearrange their environment and find visual cues to remind them to use cessation services, alongside *counterconditioning* to replace the smoking habit.

The intervention yielded a highly successful cessation prevalence of 31.5% after 6 months. However, this dropped to 10.7% after 4 years, highlighting the extreme difficulty of maintaining the final TTM stage and preventing relapse over longitudinal periods.

Case Study 2: Transport Behaviour and the Quality of Evaluation

While the TTM is highly regarded theoretically, its misapplication in the field is alarmingly common. An integrative literature review analysed 53 papers (narrowed to 13 relevant studies) applying the TTM to transport behaviour research, such as promoting public transit or active commuting to reduce carbon emissions.

The review exposed significant methodological shortcomings across the discipline. It found that although empirical studies proudly claimed to be

theoretically grounded in the TTM, the actual interventions were implemented universally, irrespective of the individual subjects' actual stages of change.

This failure to execute true "stage-matched" interventions fundamentally undermine the model's utility. The findings underscore that the TTM's analytical power lies strictly in its tailored approach to communication; applying action-stage messaging to a pre-contemplative audience is theoretically unsound and practically ineffective.

5.0 Social Cognitive Theory (SCT)

Developed by Albert Bandura, Social Cognitive Theory (SCT) transitions away from the purely internal cognitive assessments seen in the HBM or TPB, moving toward an interpersonal ecological level.

SCT emphasises the profound, continuous, and dynamic interactions between an individual's internal cognition and their external environment. Grounded in early psychological experiments regarding observational learning (such as the famous Bobo doll experiment), SCT posits that humans do not operate in a vacuum; social context is paramount.

5.1 Core Constructs and Reciprocal Determinism

At the very heart of SCT is the structural principle of **Reciprocal Determinism**. This construct dictates that personal factors (cognition, biology, mood, self-efficacy), the environment (social networks, physical surroundings, cultural and systemic structures), and human behaviour continuously interact.

They influence and are simultaneously influenced by one another in a perpetual, triadic feedback loop. A change in the environment alters behaviour, which in turn alters cognitive beliefs, which then dictates future environmental interactions.

Other critical, actionable mechanisms within SCT include:

- **Outcome Expectations**: The specific beliefs regarding the likelihood and inherent value of the consequences that will result from a behavioural choice. If the anticipated outcome is highly valued, motivation to perform the behaviour increases.

- **Self-Efficacy**: Perhaps the most widely adopted construct from Bandura's work, defined as the task-specific confidence an individual holds regarding their capability to execute a given behaviour under specific circumstances.

Importantly, self-efficacy is not synonymous with self-esteem (a general personality trait); it fluctuates heavily based on the specific task at hand.

- **Observational Learning**: The process through which individuals learn, adopt, and replicate behaviours simply by observing the actions of others (role models or peers) and witnessing the consequences those actions produce.

5.2 Empirical Case Studies in SBCC

Case Study 1: HIV Seropositivity Disclosure

The application of SCT in highly sensitive, high-stakes communication scenarios was elegantly illustrated in a study examining the disclosure of HIV seropositivity to sexual partners.

The study utilised the construct of *Outcome Expectations* by measuring precise beliefs regarding trust and rejection. For instance, researchers assessed the expectation: "I fear being rejected by my sex partner(s) if I tell him/her that I am HIV positive" on a 4-point Likert scale.

Simultaneously, it measured *Self-Efficacy* by assessing task-specific confidence, breaking down the massive behaviour of "disclosure" into micro-tasks: the ability to *initiate* the topic, the *timing* of disclosure before sexual activity, and the *resilience* to handle a partner's reaction regardless of the outcome.

SBCC interventions based on this data applied SCT by actively demonstrating the positive outcomes of disclosure (to alter negative outcome expectations) and by systematically building self-efficacy through role-playing the micro-steps of communication.

Case Study 2: Primary Care Health Promotion

A massive scoping review of 39 articles evaluated the integration of SCT in primary care health promotion interventions, revealing its dominance in clinical behaviour change. Most of these studies (including 26 randomised control trials primarily conducted in the United States) identified self-efficacy as the single most critical construct utilised to determine how behaviour change operates.

Furthermore, interventions heavily leveraged the *observational learning* construct by deploying peer group-based counselling and telephonic health coaching by specialists, essentially providing accessible social models for patients. The integration of these constructs yielded overwhelmingly positive health outcomes across diverse domains, including increased self-reported

moderate-to-vigorous physical activity, adherence to post-transplant medication regimens, and reduced high-risk STI transmission behaviours.

Case Study 3: Physical Exercise and Accelerometer Validation

Investigations testing the role of SCT components in the context of physical activity have yielded highly specific results. In studies utilising objective measures of physical activity, such as motion detectors and accelerometers worn for seven days, self-efficacy was validated as a major instigating force.

Research showed that only self-efficacy was *directly* associated with physical behaviour at a 12-month follow-up. Outcome expectancies and social support, while important, were related to behavioural goals only *indirectly*, functioning through their ability to boost self-efficacy. This reinforces that SBCC campaigns promoting exercise must prioritise building a participant's confidence to overcome barriers above all other messages.

6.0 Protection Motivation Theory (PMT)

Protection Motivation Theory (PMT), developed by R.W. Rogers in the 1970s, was originally conceptualised to explain how individuals cognitively process fear appeals.

Fear appeals are persuasive communication messages designed to intentionally scare an audience by vividly describing the terrible consequences that will occur if they do not adopt the message's recommended behaviour. PMT suggests that individuals protect themselves based on two distinct, competing cognitive appraisal processes: Threat Appraisal and Coping Appraisal.

6.1 Core Mechanisms of Appraisal

The model dictates that communicators must carefully balance fear-inducing information with empowering, actionable solutions. If threat appraisal is high but coping appraisal is low, individuals often resort to maladaptive coping mechanisms, such as denial, avoidance, or fatalism, rendering the campaign ineffective or even counterproductive.

1. **Threat Appraisal**: An evaluation of the danger itself. This encompasses:
    - *Perceived Vulnerability*: The individual's belief regarding the statistical likelihood that the specific threat will affect them directly.
    - *Perceived Severity*: The belief about how terrible the medical, social, or financial consequences of the threat will be.

- *Intrinsic/Extrinsic Rewards*: The psychological, physical, or social benefits gained by *not* changing the risky behaviour (e.g., the social reward of smoking, the physical thrill of extreme sports).

2. **Coping Appraisal**: An evaluation of the recommended preventive behaviour. This encompasses:
    - *Response Efficacy*: The logical belief that the recommended behaviour will successfully avert the threat.
    - *Self-Efficacy*: The belief in one's own ability to execute the recommended behaviour.
    - *Response Costs*: The financial, physical, or psychological costs associated with adopting the new protective behaviour.

Studies have shown that while threat appraisals initially grab attention, the coping appraisal components of PMT generally hold greater explanatory and predictive value for actual behaviour change.

6.2 Empirical Case Studies in SBCC

Case Study 1: Schistosomiasis Prevention in Rural China [*Schistosomiasis is a disease caused by parasitic worms that live in freshwater; people get infected when the larvae penetrate their skin during contact with contaminated rivers, lakes, or ponds. It mainly affects communities in tropical regions, especially in Africa and Asia, and can lead to long-term organ damage if untreated.***]**

A highly detailed predictive study of middle school students in rural China applied PMT to predict intentions to engage in protective behaviours against schistosomiasis infection. Using a stratified cluster sampling method and advanced structural equation modelling (SEM), researchers uncovered a fascinating cognitive conflict.

While 70% of participants were aware of the disease, a significant subset reported prior exposure to snail-conditioned water. The SEM analysis revealed that *intrinsic rewards* (the physical and psychological joy and recreation of playing in the water) strongly competed with the *perceived severity* of the disease.

Self-efficacy, intrinsic rewards, and perceived severity were the most significant direct predictors of behavioural intentions. Consequently, the researchers recommended that future SBCC interventions must not merely rely on situational avoidance.

Instead, they must provide realistic information about disease severity to increase the perceived threat, while simultaneously delivering intense, positive coping messages to drastically enhance self-efficacy. The students needed the confidence

to reject peer pressure and avoid the water, effectively overcoming the loss of recreational intrinsic rewards.

Case Study 2: Cognitive Behavioural Interventions for Paediatric Atopic Diseases [*Atopic diseases are a group of allergic conditions—like asthma, eczema, hay fever, and food allergies—that arise when the immune system overreacts to harmless substances such as pollen, dust, or certain foods***.]**

In a contemporary digital application of the theory, researchers in Hangzhou, China, developed a PMT-based cognitive behavioural online (PMT-CBO) intervention to manage paediatric atopic diseases.

Administered to caregivers via a WeChat mini-programme over a four-week structured course, the intervention specifically targeted the caregivers' threat appraisal and coping appraisal dimensions through 16 modules.

By unifying management strategies across the complex comorbidities of the atopic march using PMT, the online intervention successfully addressed shared pathophysiological mechanisms. Compared to a control group receiving routine verbal education, the PMT-CBO group demonstrated significantly increased protective motivation, improved preventive practices by the caregivers, and enhanced medication adherence for the children.

7.0 The Fogg Behaviour Model (FBM)

As the field of SBCC increasingly intersects with digital health, mobile applications, and user experience (UX) design, traditional cognitive models are sometimes viewed as overly academic for rapid deployment.

In response, the Fogg Behaviour Model (FBM), developed by Dr. B.J. Fogg at Stanford University, has emerged as a highly practical, streamlined framework. The model proposes a simple, universal equation for human behaviour: **B = MAP** (Behaviour equals Motivation, Ability, and a Prompt).

7.1 Core Components

The FBM asserts a rigid threshold: for any target behaviour to occur, three specific elements must converge simultaneously. If any single element is missing or falls below the action line, the behaviour will fail:

1. **Motivation (M)**: The user's underlying psychological desire to perform the behaviour. Motivation can be driven by core human drivers: sensation (pleasure/pain), anticipation (hope/fear), or social cohesion (acceptance/rejection).

2. **Ability (A)**: The capacity to perform the behaviour easily. Crucially, the FBM emphasises that increasing ability does not necessarily mean training people or giving them new skills; rather, it means modifying the environment to make the behaviour significantly *easier* to perform. This is achieved by reducing friction, time requirements, cognitive load, or financial cost.

3. **Prompts/Triggers (P)**: The immediate cue that tells the individual to perform the behaviour at that exact moment. Without a well-timed prompt, the behaviour will not happen, regardless of extraordinarily high motivation and ability.

Recent comparative reviews of the FBM and the Hook Model in higher education gamified learning environments confirm that the FBM is exceptionally effective at stimulating motivation and setting triggers within digital learning spaces, creating highly engaging educational tools.

7.2 Empirical Case Studies in SBCC

Case Study 1: Contraceptive Social Marketing During COVID-19

When the COVID-19 pandemic induced widespread lockdowns, DKT Nigeria—a reproductive health organisation—required a rapid framework to prevent catastrophic disruptions in family planning behaviours. Because a timely programmatic response was of the essence, they bypassed complex cognitive models and applied the FBM to execute adjustments across all three constructs.

To improve *Ability* (making access easier despite lockdowns), DKT established a toll-free family planning call centre (short-code *55059*) for anonymous consultations, deployed 11 additional distribution vans to ensure "last mile" clinics remained fully stocked, and issued gift vouchers to eliminate the financial barriers caused by sudden economic hardship.

To increase *Motivation*, they utilised social media to visually assure clients of clinic safety protocols (reducing the fear of COVID-19 infection) and disseminated messaging framing family planning as a crucial tool to avoid the severe financial burden of an unplanned pregnancy during a global pandemic.

Finally, they deployed specific *Triggers* via digital campaigns, prompting couples with immediate reminders that pandemic lockdowns often lead to increased sexual activity, thus serving as a direct cue to action to maintain contraceptive adherence right at the moment of risk.

Case Study 2: Sustainable Consumption Behaviour

The FBM was similarly utilised to analyse the notorious gap between green purchase intentions and actual sustainable consumption behaviour regarding bottled drinking water.

A quantitative analysis of 228 consumers tested the hypothesis that intention alone is insufficient. The data revealed that motivation, ability, and prompts functioned as a complete mediator bridging the intention-behaviour gap.

The study empirically proved that while a consumer may hold a strong, ecologically sound intention to purchase sustainably, the actual physical behaviour only actualises when the product is made highly accessible and affordable (Ability) and point-of-sale cues are conspicuously present (Prompts).

8.0 Conclusion

Individual-level models of behaviour change provide SBCC practitioners, researchers, and public health officials with the precise diagnostic tools required to understand why individuals act in specific, often seemingly irrational, ways. Furthermore, they provide the strategic blueprints required to design interventions that are scientifically grounded rather than reliant on intuition.

The Health Belief Model and Protection Motivation Theory highlight the intricate, delicate balance between rational threat assessment, fear, and individual confidence. If an intervention elevates threat without simultaneously elevating coping appraisal and self-efficacy, it risks driving the audience into denial.

The Theory of Planned Behaviour demonstrates the profound impact that societal norms and perceived behavioural control hold over human intention, necessitating culturally contextualised interventions that leverage community dynamics, as seen in the Kenyan SWS study.

The Transtheoretical Model provides a rigid temporal architecture, serving as a critical warning to practitioners against applying action-oriented communication to audiences who are merely in the precontemplation stage.

Social Cognitive Theory broadens the perspective entirely, proving through the concept of reciprocal determinism that individual cognition cannot be isolated from environmental influence, and that observational learning through peer models is one of the most effective tools in the SBCC arsenal.

Finally, the Fogg Behaviour Model offers a highly operational, modern framework, proving that artificially inflating motivation is often far more difficult than simply reducing the friction of a behaviour (Ability) and providing timely, context-aware cues (Prompts).

Collectively, a deep understanding of these theories demands that modern SBCC campaigns move far beyond generic awareness generation. To be effective, interventions must be precision-engineered to target the exact cognitive barriers, efficacy deficits, normative pressures, and environmental frictions that govern human decision-making at the individual level.